MASTER table for Desmarestia dudresnayi

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Preface

This notebook is aiming at setting the master table for good. Desmarestia dudresnayi annotation has 33 000 entries where the sister species herbacea has only ~15 000. Here I will go step by step on how I make the master table with all the features.

Ressources

Here I am listing the files I used for nf-core pipelines.
fasta: `/ebio/abt5_projects/reference_sequences/Desmarestia/Desmarestia_dudresnayi/genome/reference/Desmarestia-dudresnayi.chr.fa` **gff**: `/ebio/abt5_projects/PhaeoChromo/data/gtf-gff-bed-files/Desmarestia-dudresnayi_BAKER.chr.AGAT.gff`

Note: this gff file is derived from the reference gtf file, using AGAT tool to fix it and make it compatible with nf-core pipelines.

Code previously used to make the master table `./code/03_MASTER_TABLE_Desmarestia-dudresnayi.R` and then to add signatures to it `07_michael_mergeCS_v3.r`.

```
# Setting up environment
library(here)  # Optional but recommended for better path management
```

```
here() starts at /home/jeromine/Documents/Scripts_Rstudio/MASTER_TABLES/Notebooks
```

```
knitr::opts_knit$set(root.dir = "/home/jeromine/Documents/Scripts_Rstudio/MASTER_TABLE
knitr::opts_chunk$set(echo = TRUE)
```

Cleaning up GFF file

```
# Loading lib ------------------------------------------------------------
library(rtracklayer)
library(dplyr)
library(dlookr)
library(tidyr)
library(stringr)
library(data.table)
library(GenomicRanges)
library(ggplot2)
library(ggridges)
```

Step 1: Load GFF File

```
# Load GFF file
gff <- import("./data/gff_files/Desmarestia-dudresnayi_BRAKER.chr.gff_sorted", format
# Convert GRanges to a data frame
gff <- as.data.frame(gff)
# Convert to data.table for fast processing
setDT(gff)
print(sum(gff$type == "mRNA"))
```

```
[1] 33160
```

Step 2: Remove Duplicated Gene Model between AUGUSTUS and GeneMark

Screenshot in IGV that shows overlaping AUGUSTUS and GeneMark model:

Screenshot of IGV showing overlapping AUGUSTUS and GeneMark model

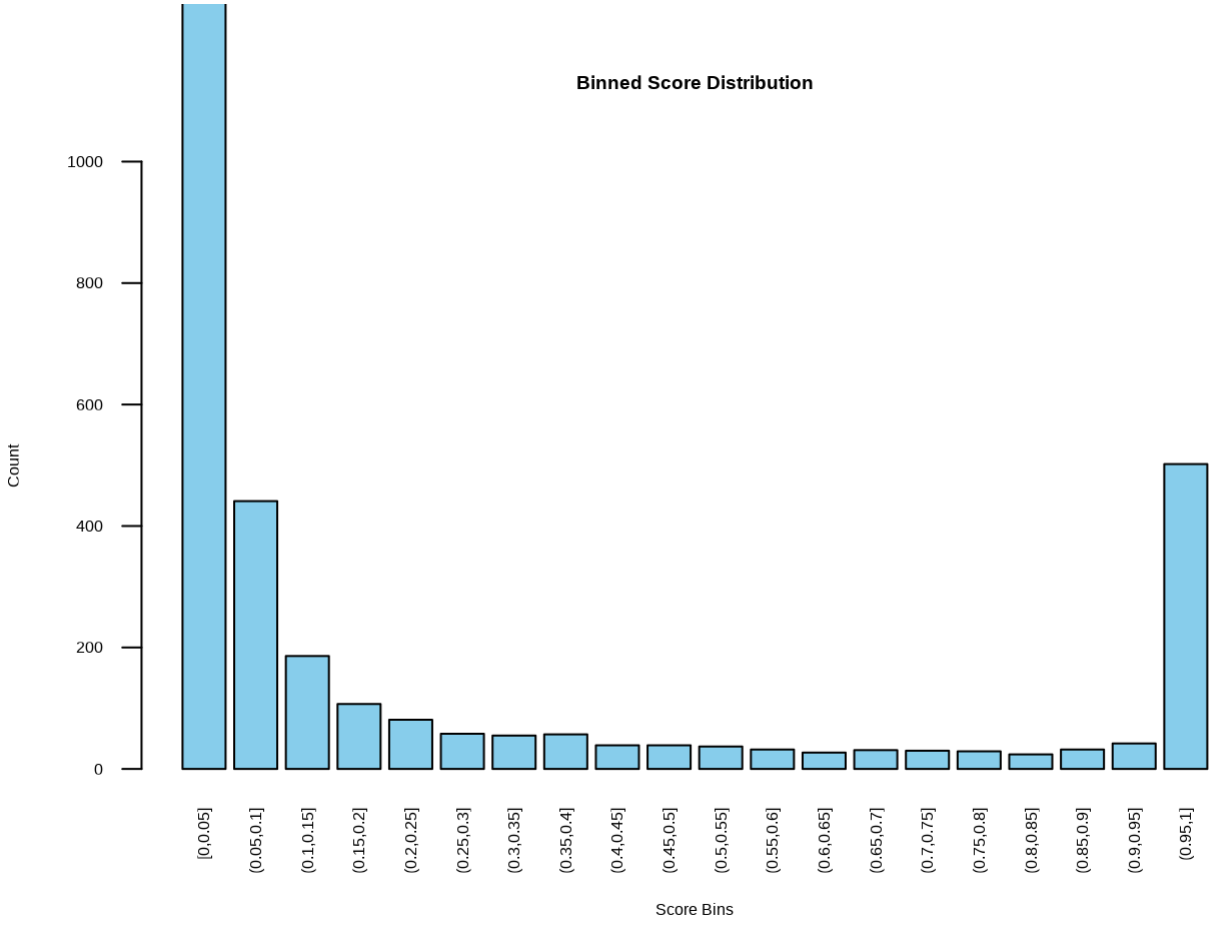
```
# Remove duplicated gene model
gff <- subset(gff, source == "AUGUSTUS")
print(sum(gff$type == "mRNA"))
```

```
[1] 31767
```

Option1 : multi-filtering

Step 3: Remove TE-Overlapping Genes

```
gff_data <- gff
CDS_TE_table <- read.table(file = './data/AGORE_TE-filtering_rcraig_for_jero/dDud.CDS-

# Define bins (adjust 'breaks' as needed for better granularity)
CDS_TE_table$score_bins <- cut(CDS_TE_table$score, breaks=seq(0, 1, by=0.05), include.

# Count occurrences in each bin
score_counts <- table(CDS_TE_table$score_bins)

# Create a barplot
barplot(score_counts, main="Binned Score Distribution", xlab="Score Bins", ylab="Count
```




```
# Filter overlap > 70%
CDS_TE_table <- CDS_TE_table %>% filter(score > 0.7)
CDS_TE_table$gene_id <- gsub('\\.t.*','',CDS_TE_table$trad_id)

# Filter out genes that are in the TE list
gff_data$GParent <- ifelse(is.na(gff_data$Parent),'',gsub('\\.t.*','',gff_data$Parent)
# Ensure Parent column does not contain 'character(0)'
gff_data$Parent <- ifelse(gff_data$Parent == "character(0)", NA, gff_data$Parent)
gff_data <- gff_data[!gff_data$GParent %chin% CDS_TE_table$gene_id, ]
gff_data <- gff_data[!gff_data$ID %chin% CDS_TE_table$gene_id, ]
print(sum(gff_data$type == "mRNA"))
```

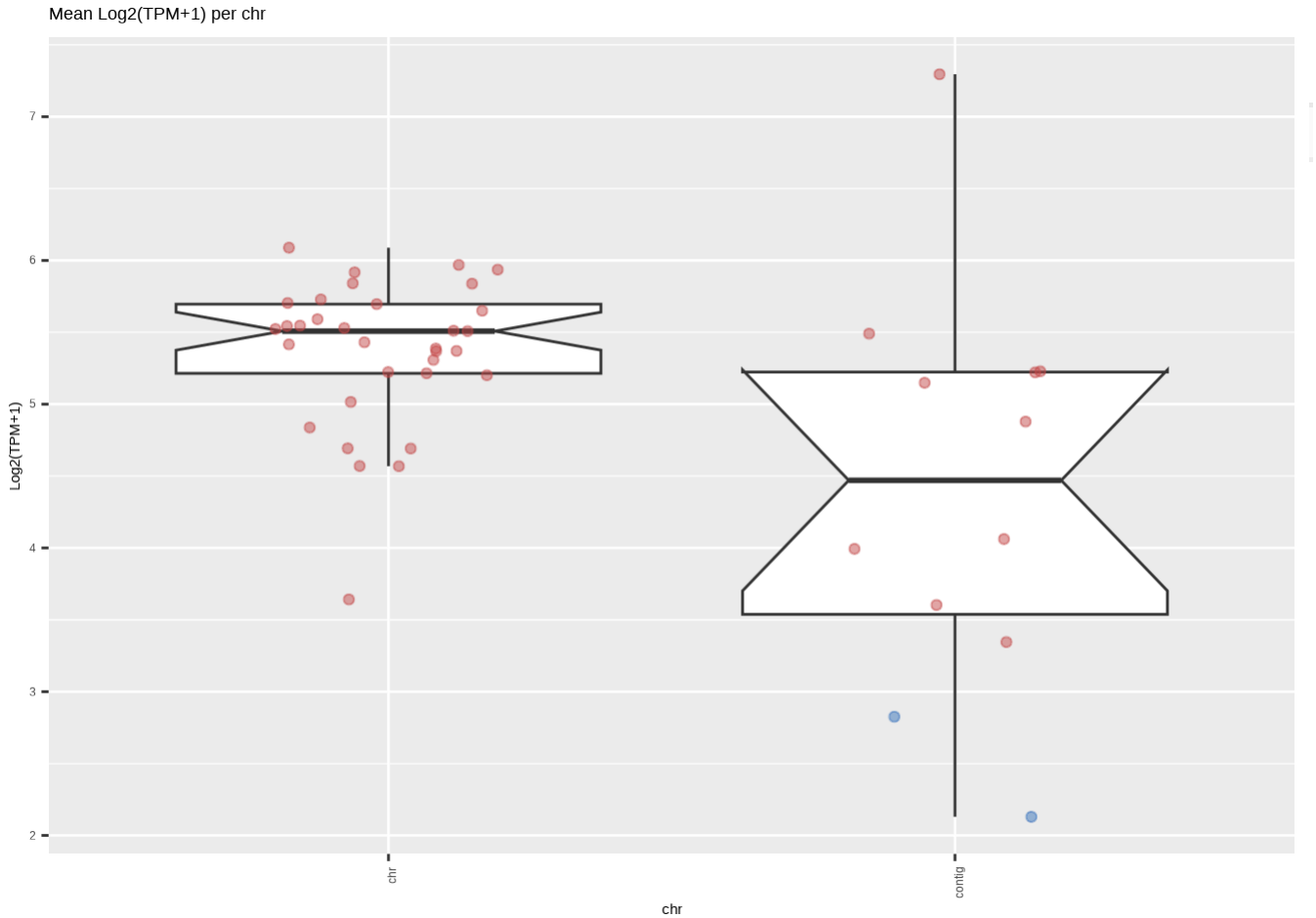
```
[1] 31103
```

Step 4: Filter Bacterial Contigs

```
## Define bacterial contigs: Bacterial contigs will be contigs where there are no read
# Load TPM data
tpm_data <- read.table('./data/my_RNAseq_data/Desmarestia-dudresnayi/salmon.merged.gen
# Calculate the TPM mean
tpm_data <- tpm_data %>%
  mutate(meanTPM = rowMeans(select(., Dd1, Dd2, Dd3), na.rm = TRUE)) %>%
  select(gene_id, meanTPM)
```

```
# Merge TPM with GFF data to get seqid
gff_tpm_data <- merge(gff_data, tpm_data, by.x = "GParent", by.y = "gene_id")

# Group by seqid, calculate mean TPM, and log-transform
tpm_summary <- gff_tpm_data %>%
  group_by(seqnames) %>%
  summarise(mean_scaffold_TPM = mean(meanTPM, na.rm = TRUE)) %>%
  mutate(log2_scaffoldTPM_plus_1 = log2(mean_scaffold_TPM + 1)) %>%
  mutate(cat = case_when(grepl("^chr", seqnames) ~ "chr",
                         grepl("^contig", seqnames) ~ "contig",
                         TRUE ~ "other"))

# Plot mean TPM per seqid
ggplot(tpm_summary, aes(x = cat, y = log2_scaffoldTPM_plus_1)) +
  geom_boxplot(outlier.shape = NA, notch = TRUE) +
  geom_jitter(aes(color = ifelse(cat == "contig" & log2_scaffoldTPM_plus_1 < 3, "#3F78
              width = 0.2, alpha = 0.5) +
  scale_color_identity() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(title = "Mean Log2(TPM+1) per chr", x = "chr", y = "Log2(TPM+1)")
```

```
Notch went outside hinges
ℹ Do you want `notch = FALSE`?
```



```
# List bacterial contigs
bacterial_contigs <- tpm_summary$seqnames[tpm_summary$cat == 'contig' & tpm_summary$lo
print(length(bacterial_contigs))
```

```
[1] 2
```

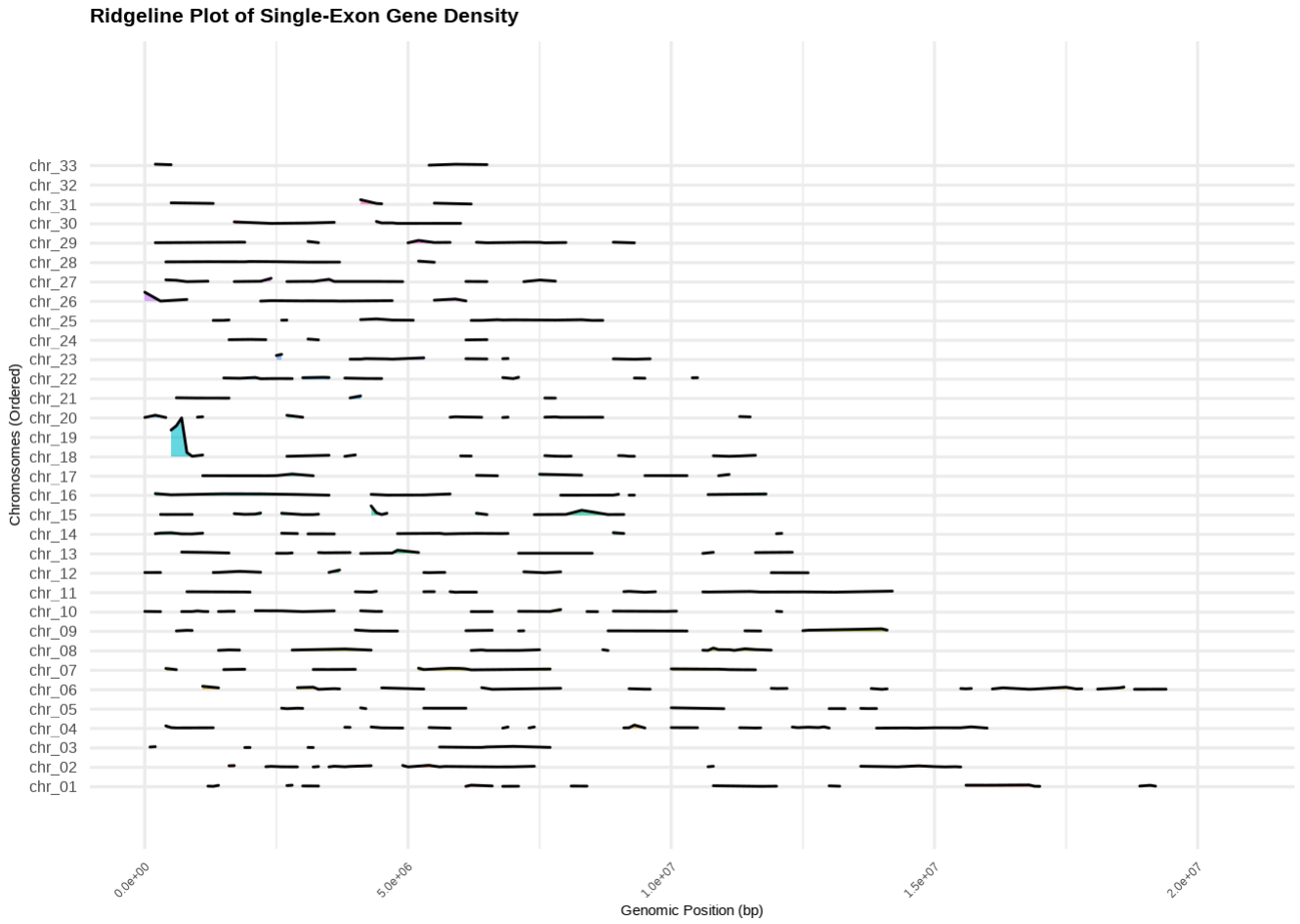
```
# Remove bacterial contigs
gff_data <- gff_data[!grepl(paste(bacterial_contigs, collapse = "|"), seqnames), ]
print(sum(gff_data$type == "mRNA"))
```

```
[1] 31097
```

Step 5: Remove Monoexonic Bacterial Genes

```
# Identify exon entries
exon_data <- gff_data[gff_data$type == "exon"]

# Count exons per gene using dplyr
gene_exon_counts <- exon_data %>%
  as.data.frame() %>%
  group_by(Parent) %>%
  summarise(exon_count = n(), .groups = "drop")
gene_exon_counts <- gene_exon_counts %>%
  mutate(GParent = gsub('\\.t.*','',gene_exon_counts$Parent))

# Shortlist monoexonic genes
single_exon_genes <- gene_exon_counts %>%
  filter(exon_count == 1)

# Filter genes with only one exon
single_exon_data <- exon_data[exon_data$GParent %chin% single_exon_genes$GParent]

# Define window size (e.g., 100,000 base pairs)
window_size <- 100000
# Compute monoexon density
single_exon_density <- single_exon_data %>%
  mutate(window = floor(start / window_size) * window_size) %>%  # Assign window bin
  group_by(seqnames, window) %>%
  summarise(SE_bp = sum(width), .groups = "drop") %>%  # Sum monoexon base-pairs per w
  mutate(SE_density = SE_bp / window_size)  # Compute density

# Ensure seqnames (chromosome names) are properly formatted
single_exon_density_toplot <- single_exon_density %>%
  filter(grepl("^chr", seqnames))  # Keep only seqnames that start with "chr"
write.csv(single_exon_density, './Notebooks/single-exon-density.csv', quote = FALSE, r

# Ridgeline plot with increased spacing and better scaling
ggplot(single_exon_density_toplot, aes(x = window, y = seqnames, height = SE_density,
  geom_density_ridges(stat = "identity", alpha = 0.6, scale = 2, rel_min_height = 0.01
  labs(title = "Ridgeline Plot of Single-Exon Gene Density",
```

```
       x = "Genomic Position (bp)",
       y = "Chromosomes (Ordered)") +
  scale_y_discrete(expand = expansion(mult = c(0.1, 0.2))) +  # Ensures spacing
  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 12),
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(size = 16, face = "bold"),
    legend.position = "none"
  ) +
  expand_limits(y = 0)  # Ensures taller rows don't get cut off
```

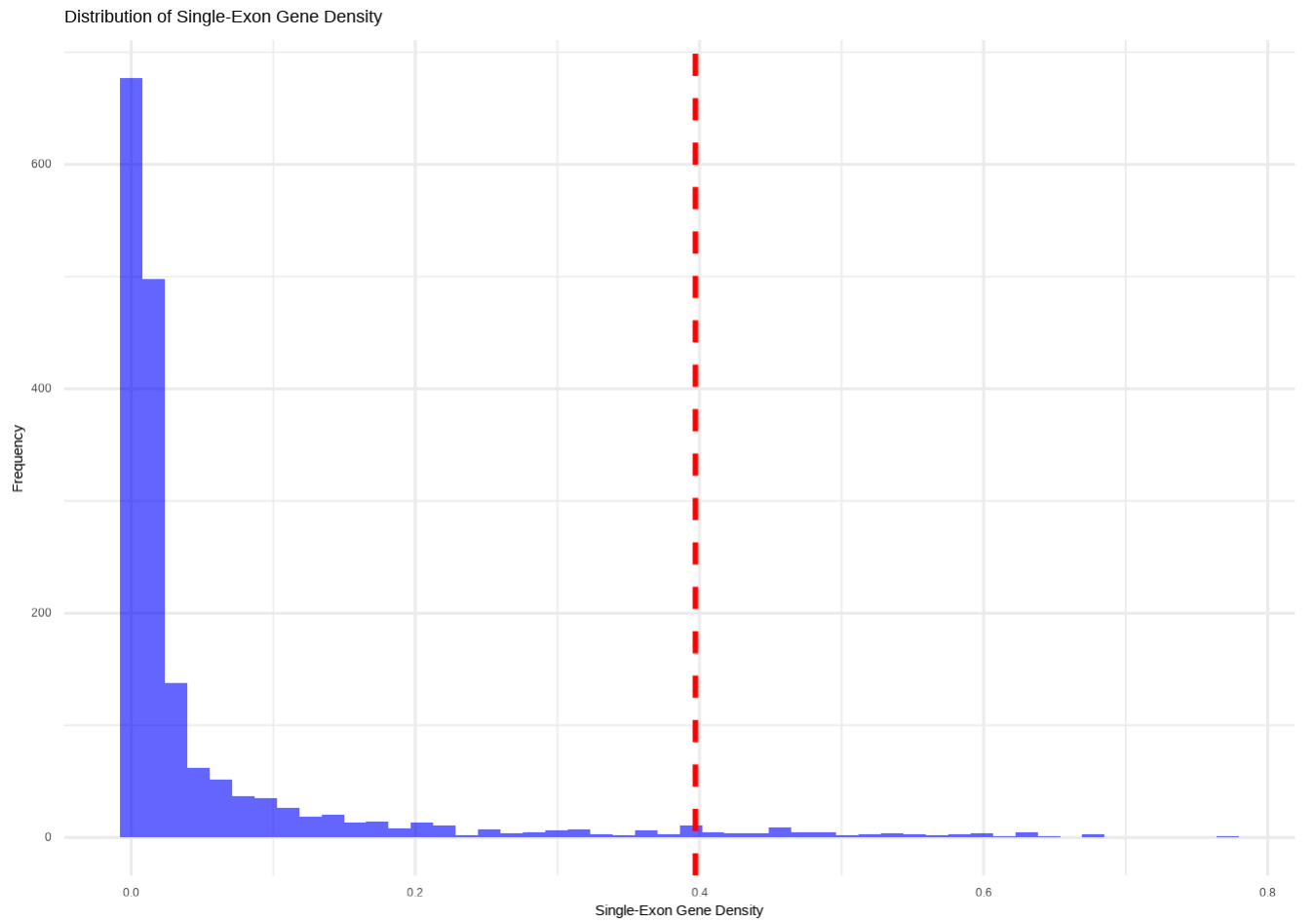



```
# Save a bigger version
ggsave("./Notebooks/monoexon_density_ridgeline_tall.png", width = 14, height = 15, dpi

# Compute the mean and standard deviation of SE_density
mean_density <- mean(single_exon_density$SE_density, na.rm = TRUE)
std_density <- sd(single_exon_density$SE_density, na.rm = TRUE)

# Define the Z-score threshold (mean + 3*std)
z_threshold <- mean_density + 3 * std_density

# Identify regions above the threshold
contaminated_regions <- single_exon_density[single_exon_density$SE_density > z_thresho
  select(seqnames, window, SE_bp)
```

```
# Convert to BED format (0-based start, end = window + SE_bp)
contaminated_regions <- contaminated_regions %>%
  mutate(start = window,
         end = window + SE_bp) %>%
  select(seqnames, start, end)

# Print the threshold
print(paste("Z-score threshold:", round(z_threshold, 5)))
```

```
[1] "Z-score threshold: 0.39715"
```

```
# Plot the distribution of SE_density with the threshold
ggplot(single_exon_density, aes(x = SE_density)) +
  geom_histogram(bins = 50, fill = "blue", alpha = 0.6) +
  geom_vline(xintercept = z_threshold, color = "red", linetype = "dashed", linewidth =
  labs(title = "Distribution of Single-Exon Gene Density",
       x = "Single-Exon Gene Density",
       y = "Frequency") +
  theme_minimal()
```




```
# Filter out monoexonic genes
# Convert to GRanges objects
gff_data <- gff_data %>%
```

```
  mutate(attributes = paste0("ID=", ID, ifelse(!is.na(Parent), paste0(";Parent=", Pare
  mutate(score = ifelse(is.na(score), '.', score)) %>%
  mutate(phase = ifelse(is.na(phase), '.', phase)) %>%
  select(seqnames,source,type,start,end,score,strand,phase,attributes)
gff_gr <- GRanges(seqnames = gff_data$seqnames, ranges = IRanges(start = gff_data$star
contaminated_gr <- GRanges(seqnames = contaminated_regions$seqnames, ranges = IRanges(

# Find overlaps
overlaps <- findOverlaps(gff_gr, contaminated_gr, type = "within")

# Remove contaminated features
clean_gff <- gff_data[-queryHits(overlaps), ]

# Save the masked GFF file
write.table(clean_gff, "./Notebooks/Dd_masked_output.gff", sep = "\t", quote = FALSE,
clean_gff <- as.data.table(clean_gff)

print(sum(clean_gff$type == "mRNA"))
```

```
[1] 28756
```

Option2 : Keeping orthologous genes from herbacea only

```
# Convert GRanges to a data frame
gff_data2 <- as.data.frame(gff)
# Convert to data.table for fast processing
setDT(gff_data2)
gff_data2$GParent <- ifelse(is.na(gff_data2$Parent),'',gsub('\\.t.*','',gff_data2$Pare

OG <- read.table('./data/Orthofinder/Results_Dec12/Orthogroups/Orthogroups.tsv', heade
rownames(OG) <- OG[,1]
OG <- OG %>%
  select(Desmarestia.dudresnayi_BRAKER_proteins, Desmarestia.herbacea_MALE_proteins) %
  filter(!is.na(Desmarestia.herbacea_MALE_proteins) & Desmarestia.herbacea_MALE_protei
OG <- OG %>%
  select(Desmarestia.dudresnayi_BRAKER_proteins) %>%
  separate_rows(Desmarestia.dudresnayi_BRAKER_proteins, sep = ",") %>%
  rename(trad_id = Desmarestia.dudresnayi_BRAKER_proteins)
OG$gene_id <- gsub('\\.t.*','',OG$trad_id)

gff_data2 <- gff_data2[gff_data2$GParent %chin% OG$gene_id, ]

gff_data2 <- gff_data2 %>%
  mutate(attributes = paste0("ID=", ID, ifelse(!is.na(Parent), paste0(";Parent=", Pare
  mutate(score = ifelse(is.na(score), '.', score)) %>%
  mutate(phase = ifelse(is.na(phase), '.', phase)) %>%
  select(seqnames,source,type,start,end,score,strand,phase,attributes)

print(sum(gff_data2$type == "mRNA"))
```

```
[1] 10641
```

Step 7: Export Cleaned GFF File

```
# Write new gff file

# Write few comments
cat('# This file was generated from reference gtf Desmarestia-dudresnayi, then convert
   # 1 - Removing duplicated gene model, 2 - Removing TE genes, 3 - Removing contigs p
file = "./data/gff_files/Desmarestia-dudresnayi.cleaned2025.jv.gff")
write.table(clean_gff, file = "./data/gff_files/Desmarestia-dudresnayi.cleaned2025.jv.

cat('# This file was generated from reference gtf Desmarestia-dudresnayi, then convert
   # 1 - Removing duplicated gene model, 2 - Keeping Desmarestia herbacea orthologous
file = "./data/gff_files/Desmarestia-dudresnayi.cleaned2025_OGonly.jv.gff")
write.table(gff_data2, file = "./data/gff_files/Desmarestia-dudresnayi.cleaned2025_OGo
```

MASTER TABLE

see next notebook